

How Fuel Subsidy Reductions Reshaped Fishing Activity: Evidence from China's Policy Reform in the East China Sea

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Abstract

This paper examines how fuel subsidy reductions affect fishing activity, exploiting the 2016 reform of China's fuel subsidy program in the East China Sea as a quasi-natural experiment. The reform reduced average subsidy payments by 69% and introduced an asymmetric distribution across vessel sizes, with larger vessels receiving 15.8% more support than smaller ones. We develop a theoretical framework to characterize fishers' responses to differential subsidy cuts and test its predictions using Regression Discontinuity Design and Difference-in-Differences approaches. Our results show that while total fishing effort declined following the reform, unauthorized fishing increased by 19.3%, particularly in Taiwanese waters and disputed maritime zones. We find that larger vessels—which retained higher subsidy intensity—drove the expansion of unauthorized activity, whereas smaller vessels adjusted along the intensive margin over time. Consistent with our model, vessels facing larger relative subsidy reductions were 2.5% more likely to exit the sector, while those that remained intensified their fishing effort per unit of subsidy. These findings highlight a significant *effort leakage* across space and legal regimes, suggesting that subsidy reforms, while effective in lowering aggregate domestic effort, may induce unintended shifts toward unauthorized fishing and exacerbate geopolitical tensions.

Keywords: Fisheries Management, Fuel Subsidies, Unauthorized Fishing, Policy Evaluation, China.

1 Introduction

Input subsidies have been widely used as instruments to enhance productive capacity across sectors (Reyes-García et al. 2025). In fisheries, such subsidies have been linked to overcapacity and the overexploitation of marine resources, particularly in open-access settings (Munro and Sumaila 2002; Sumaila et al. 2021; Shen and Chen 2022). In 2018, global fisheries subsidies amounted to USD 35.4 billion, of which USD 22.2 billion were capacity-enhancing (Sumaila et al. 2019). These policies have contributed to unsustainable fishing practices, especially in contexts with weak enforcement (Sumaila et al. 2010a; Wang et al. 2023; Skerritt et al. 2023).

According to FAO (2024), the share of fish stocks outside biologically sustainable levels has increased, signaling accelerating marine ecosystem degradation. Capacity-enhancing subsidies relax cost constraints, enabling vessels to operate beyond national waters (Englander et al. 2025) and generating profits that would not arise otherwise (Sumaila and Pauly 2013; Sumaila et al. 2019; Sakai et al. 2019). These policies also distort public budgets, raising the opportunity cost for governments and society (Davis 2014), and contribute to broader social externalities, including piracy (Denton and Harris 2021; Phayal et al. 2024; Mitchell and Schmidt 2024). In response, the World Trade Organization (WTO) has advanced negotiations to eliminate such subsidies to curb overfishing and achieve Sustainable Development Goal (SDG) 14.6 (Yang et al. 2017; Skerritt and Sumaila 2021; Cisneros-Montemayor et al. 2022; Shen and Chen 2022).

Despite growing causal evidence on how subsidies affect fishing capacity and overfishing (Shen and Chen 2022; Wang et al. 2023; Englander et al. 2025), there is limited empirical evidence on how subsidy reductions shape incentives for Illegal, Unreported, and Unregulated (IUU) fishing and how vessels adjust behavior when facing differential policy changes.

This paper studies how reductions in fuel subsidies affect fishing effort and unauthorized fishing¹ behavior by Zhejiang Province’s industrial fishing fleet, particularly in ecologically and geopolitically sensitive waters such as those of Taiwan, Japan, and Disputed Zones². We exploit the 2016 Chinese reform, which introduced asymmetric reductions in fuel subsidies across vessels based on their length—specifically, around the 30-meter threshold where subsidy allocation change discontinuously—, together with the Vessel Buyback Program (VBP), which increased exit incentives. We develop a theoretical model that links subsidy changes to the allocation of authorized and unauthorized effort, and test its predictions using vessel-level data, and a Regression

¹Unauthorized fishing refers to vessels operating in foreign waters without formal access agreements, which constitutes one dimension of IUU fishing (FAO 2024).

²Disputed zones are usually maritime areas claimed by two or more countries where their Exclusive Economic Zones (EEZs) overlap. They often lead to tensions as each nation asserts its own sovereign rights and, therefore, the right to exploit the disputed territory (Østhagen 2020).

Discontinuity Design (RDD) and Difference-in-Differences (DiD) strategies combining administrative records, Global Fishing Watch (GFW) data, and information on fishing access agreements.

We focus on the Chinese case because it is the largest provider of fisheries subsidies worldwide. In 2018, it allocated USD 7.2 billion, representing 21% of global fisheries subsidies, 81% of which were classified as capacity-enhancing (Sumaila et al. 2019). This level of support has contributed to the expansion of China’s fishing capacity, raising international concern over its impact on the sustainability of marine ecosystems (Englander et al. 2025). In addition, China’s extensive involvement in IUU fishing has become a source of diplomatic tension (Tseng and Ou 2010; Zhang 2016; Kim 2019; Watson and Woodill 2022). Zhejiang province plays a central role in this context. It hosts the largest fishing fleet in the East China Sea, with a significant share of vessels operating through trawling methods (Ministry of Agriculture Fisheries Administration Bureau 2006), one of the most destructive practices for marine ecosystems (Buhl-Mortensen and Buhl-Mortensen 2018). Furthermore, it is governed by China’s most prominent coastal authority (Wang et al. 2023), making it a strategically important area for examining the effects of subsidy reductions on fishing activities, including IUU fishing in Disputed Zones and the surrounding regions.

Our results show that total fishing effort declines after the reform, particularly among smaller vessels. This pattern reflects both their response to lower subsidy provision (15.8%) and the decline in fishing capacity driven by the VBP (2.5%). On the other hand, unauthorized fishing increases, especially among larger vessels operating in disputed zones. We also find that vessels adjust their effort-to-subsidy ratios³, suggesting compensatory behavior in response to reduced financial support.

This paper contributes to the literature on the effects of input subsidies, specifically the impacts of differential reductions in subsidies payments (Sumaila et al. 2010a, 2021; Wang et al. 2023; Englander et al. 2025), in three ways. First, it provides causal evidence on how subsidy reductions affect not only fishing effort but also unauthorized fishing behavior. Second, it highlights the role of heterogeneous policy exposure in shaping behavioral responses across vessel types. Third, it shows that subsidy reforms, in the absence of complementary enforcement, may generate unintended incentives toward unauthorized fishing.

The remainder of the paper is structured as follows. Section 2 provides a description of the case study, including background information and the institutional context of the fuel subsidy reform. Section 3 introduces the theoretical model. Section 4 describes the data and the empirical strategy. Section 5 presents the results on the effects of the reform. Section 6 discusses the findings and concludes.

³The effort-to-subsidy ratio corresponds to fishing effort (in hours) divided by fuel subsidies (measured in RMB/kW, i.e., Chinese yuan per unit of engine power in kilowatts), and provides a normalized measure of effort relative to subsidy support.

2 Study Case

2.1 Background

The East China Sea has long been a region of territorial disputes over sovereignty and access to marine resources (Valencia 2014; Midford and Østhagen 2024). In this context, Chinese fishers have become central actors in maritime tensions, as government policies have directly and indirectly shaped their presence in disputed zones (Zhang 2016). These dynamics are further complicated by the divergence between China's maritime claims and those recognized under international law, contributing to the persistent presence of Chinese vessels in contested areas (Radio Free Asia 2024).

The expansion of Chinese fishing fleets has intensified concerns over overexploitation and unauthorized fishing, particularly in the Sea of Japan. Between 2018 and 2020, Chinese vessels were detected on average 4,341 times within Japanese territorial waters, with a high concentration in disputed areas around the Diaoyu/Senkaku Islands (Sato and Chadha 2022; Reuters 2024). Part of this activity reflects transit toward North Korean waters, where fishing agreements exist, highlighting the interaction between geopolitical constraints and fishing behavior. More recently, large-scale coordinated deployments of Chinese fishing vessels have further underscored the strategic dimension of these activities, reinforcing concerns about their role in maritime disputes and state-backed presence in contested waters ⁴. Tensions are also evident in Taiwanese waters, especially around Kinmen, located approximately 4 km from the Chinese coast (Radio Free Asia 2024). Figure 1 illustrates these maritime boundaries.

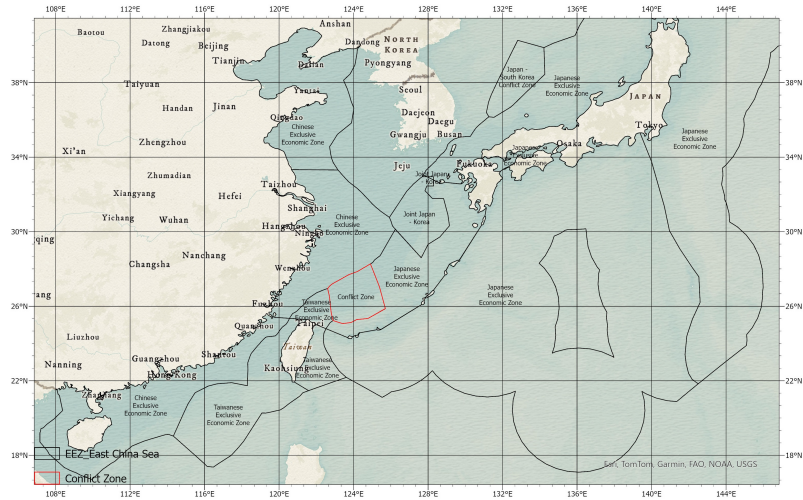


Fig. 1: Geographic scope of the study. Map of the East China Sea showing the Exclusive Economic Zones (EEZs) of China, Japan, and Taiwan, as well as disputed maritime areas. The black solid lines represent EEZ boundaries, while the red line indicates disputed zones. The map highlights the main regions where Chinese industrial vessels operate and where unauthorized fishing activity is analyzed. Source: Authors, based on [Marine Regions Repository](#).

⁴See: <https://www.nytimes.com/interactive/2026/01/16/world/asia/china-ships-fishing-militia-blockade.html>

Institutionally, there are no formal fishing access agreements between China, Japan, and Taiwan, largely due to concerns over resource sustainability and long-standing diplomatic frictions (Chen et al. 2024). In contrast, Taiwan and Japan signed a bilateral fisheries agreement in 2013 allowing reciprocal access in overlapping EEZs, effectively excluding China from cooperative management arrangements in the region (The Diplomat 2018). This asymmetry further shapes strategic interactions in contested waters.

The presence of Chinese vessels in foreign waters is primarily driven by structural overcapacity (Wang et al. 2023) and policy objectives related to food security (Zhang 2016). Combined with subsidy provision, these factors incentivize greater fishing effort and might increase the likelihood of operating outside regulated frameworks. This combination of geopolitical tensions, lack of access agreements, and persistent subsidies makes the East China Sea a particularly relevant setting to examine how changes in subsidy schemes affect incentives for unauthorized fishing.

2.2 Institutional and Policy Overview

China plays a central role in global fisheries due to its scale of operations and fishing capacity, the largest worldwide (Shen and Heino 2014). Despite regulatory efforts since 1979, capacity has continued to expand, contributing to the widespread depletion of domestic fish stocks (Yu and Yu 2008; Teh et al. 2017; Zhang et al. 2025). Between 2012 and 2014 alone, capacity increased by 149% (Yang et al. 2017). This context provides a unique setting to study how capacity-enhancing subsidies shape fishing behavior in response to changing cost structures and policy incentives, which we examine through the 2016 fuel subsidy reform and the Vessel Buyback Program (VBP), targeting operating costs and fleet capacity and generating clear empirical variation.

2.2.1 Fuel subsidy program

The expansion of China’s fishing capacity has been closely linked to input subsidies (Yang et al. 2017; Mallory 2016). Following a partial liberalization of fuel prices in 2006, the government introduced targeted subsidies for fisheries based on vessel and operational characteristics (Englander et al. 2025).

In 2016, the program was reformed in response to environmental concerns and fiscal pressures (Su et al. 2020). The reform introduced a classification system based on vessel size and gear type, reduced the baseline fuel price standard, and adjusted subsidy coefficients to penalize environmentally harmful practices. Importantly, it generated discontinuities in subsidy allocation, particularly at the 30-meter vessel length threshold for trawlers (Wang et al. 2023). Following the 2016 reform, fuel subsidy payments experienced a sharp decline of approximately 69%. Crucially, this reduction was asymmetric, as larger vessels received more subsidies than smaller ones, creating the discontinuity used in our empirical design (see Figure S1 in the supplementary material).

2.2.2 Vessel buyback program

In parallel, China has implemented a VBP since 2002, aimed at reducing fleet capacity by incentivizing exit from the sector (Graff Zivin and Mullins 2015). The program targets high-impact vessels, such as trawlers, and operates alongside a licensing system that restricts entry by requiring new vessels to acquire licenses from decommissioned ones (Wang et al. 2023; Su et al. 2020). Between 2002 and 2014, the VBP reduced the number of commercial vessels by 31% (Cao et al. 2017).

However, its effectiveness was initially limited by the coexistence of fuel subsidies that incentivized continued participation (Cao et al. 2017). The 2016 reform altered this balance: reductions in subsidy expenditures allowed for increased buyback incentives, particularly in Zhejiang Province (Wang et al. 2023). As a result, exit rates rose, especially among smaller vessels, which faced larger subsidy cuts. Figure S2 in the supplementary material shows that vessels below 30 meters exhibited a higher exit rate in the post-reform period (2016-2019).

Together, the subsidy reform and the VBP provide a setting to analyze how changes in cost structures and opportunity costs affect fishing behavior. While reducing subsidies is expected to lower overcapacity and overexploitation (Englander et al. 2025; Wang et al. 2023), its effect on unauthorized fishing remains theoretically ambiguous, making it essential to examine behavioral responses.

3 The Model

Consider a fishery with a large number of vessels $i = 1, \dots, n$, each of type $t_i \in \{s, l\}$ (small or large). Let $S_i \in \mathbb{R}_+$ denote the subsidy received by vessel i . Prior to the reform, subsidies were broadly similar across vessel types. A reform r reduces subsidies unevenly across types, such that

$$S_i^r = (1 - \lambda_{t_i})S_i. \quad (1)$$

Where $0 < \lambda_l < \lambda_s < 1$. Each vessel chooses effort to maximize payoffs, allocating total effort $e_i = e_{a_i} + e_{na_i} \in \mathbb{R}_+$ between authorized and unauthorized fishing. Benefits are given by a revenue function $b_i : \mathbb{R}_+ \rightarrow \mathbb{R}_+$, increasing and concave in effort, while costs are constant at c_{a_i} and c_{na_i} for authorized and unauthorized fishing, respectively.⁵ Payoffs depend on others' aggregate effort $E_{-i} = \sum_{j \neq i} e_j$, which generates a negative externality $N_i(E_{-i}) \in \mathbb{R}_+$, increasing in E_{-i} . Assuming payoff symmetry across vessels,⁶ payoffs are defined as

$$\pi_i(e_{a_i}, e_{na_i}; E_{-i}) = b_i(e_{a_i} + e_{na_i}) - (c_{a_i} - S_i)e_{a_i} - c_{na_i}e_{na_i} - N_i(E_{-i}). \quad (2)$$

Maximizing vessel payoffs yields the optimal allocation of effort and the corresponding first-order condition for authorized effort

⁵Effort is measured in standardized units (e.g., trips or fishing days), implying constant marginal input requirements.

⁶Although larger vessels may achieve higher profits due to scale or technology, introducing such heterogeneity would not affect the main qualitative insights of the model. We abstract from these differences and consider economically homogeneous vessels for simplicity.

$$\frac{\partial \pi_i(e_{a_i}, e_{na_i}; E_{-i})}{\partial e_{a_i}} = \frac{\partial b_i(e_{a_i} + e_{na_i})}{\partial e_{a_i}} - (c_{a_i} - S_i) = 0. \quad (3)$$

This condition implies that, at the optimum,

$$\frac{\partial b_i(e_{a_i} + e_{na_i})}{\partial e_{a_i}} = c_{a_i} - S_i. \quad (4)$$

In (4), the subsidy lowers the marginal cost of authorized fishing, increasing incentives to allocate effort to authorized activities. Combining (1) and (4), the following proposition characterizes how the subsidy reform affects authorized effort.

Proposition 1. *A reduction in subsidies increases the marginal cost of authorized fishing, reducing effort. As small vessels experience proportionally larger cuts, their decline in effort may be more pronounced.*

Proof: See Section 4.1 in the supplementary material.

Considering this adjustment in authorized fishing, the optimal allocation of unauthorized fishing is determined by the first-order condition

$$\frac{\partial \pi_i(e_{a_i}, e_{na_i}; E_{-i})}{\partial e_{na_i}} = \frac{\partial b_i(e_{a_i} + e_{na_i})}{\partial e_{na_i}} - c_{na_i} = 0. \quad (5)$$

Authorized fishing typically entails higher compliance and regulatory costs, raising its marginal cost relative to unauthorized fishing. Subsidies offset this differential and incentivize legal activity, so that prior to the reform they are assumed to more than compensate these costs, making authorized fishing cheaper at the margin. Formally,

$$c_{a_i} - S_i < c_{na_i}. \quad (6)$$

The reform reduces this compensating transfer, raising the effective marginal cost of authorized fishing to $c_{a_i} - S_i^r$. If the reduction is large enough, the cost advantage of authorized fishing may be attenuated or reversed, yielding

$$c_{a_i} - S_i < c_{na_i} \geq c_{a_i} - S_i^r. \quad (7)$$

Changes in costs may alter vessels' effort allocation across fishing activities.⁷ Combining (1), (6), and (7), the following proposition characterizes the effect of the subsidy reform on unauthorized fishing.

Proposition 2. *A reduction in subsidies increases the relative cost of authorized fishing, potentially prompting vessels to shift effort toward unauthorized fishing.*

Proof: See Section 4.2 in the supplementary material.

The intuition follows from changes in relative marginal costs. If the subsidy reduction raises the cost of authorized fishing above that of unauthorized fishing, vessels may reallocate effort accordingly. Under payoff symmetry, heterogeneity arises only

⁷ A flexible interpretation of equation (7) allows for the possibility that the inequality $c_{na_i} < c_{a_i} - S_i^r$ reflects either the actual economic cost structure or the behavioral response of fishers to the subsidy reform. In other words, after the reform, the higher net cost of authorized fishing allows fishers to perceive unauthorized fishing as relatively more attractive. This perception may arise from changes in expected revenue, higher opportunity costs of authorized fishing, or imperfect information about enforcement and profitability. Thus, the inequality allows the model to capture both direct cost effects and strategic adjustments in effort allocation under the new subsidy regime.

from differences in subsidy cuts, with smaller vessels—facing larger proportional reductions—more likely to shift toward unauthorized fishing. Allowing for cost heterogeneity introduces an additional channel: if unauthorized fishing is relatively cheaper for larger vessels, they may increase unauthorized effort more strongly. As a result, heterogeneous responses are theoretically ambiguous and depend on the joint distribution of subsidy reductions and cost differences across vessel types, requiring empirical assessment.⁸

The model also provides implications for the overall level of fishing activity.

Proposition 3. *A reduction in subsidies is expected to lower total fishing effort, with small vessels potentially reducing effort more due to proportionally larger cuts.*

Proof: See Section 4.3 in the supplementary material.

Although some vessels may increase unauthorized fishing, reductions in authorized fishing are not fully offset because authorized and unauthorized fishing are imperfect substitutes. As a result, total fishing effort is expected to decline.

Changes in the cost structure may affect vessel participation in the fishery by influencing exit decisions through the vessel buyback program.

Proposition 4. *A reduction in subsidies may induce less competitive vessels to exit the fishery, particularly when exit incentives such as the vessel buyback program are available.*

Proof: See Section 4.4 in the supplementary material.

In this context, higher operating costs reduce fishing profitability, increasing incentives to exit in a setting where the subsidy reform coincide with a vessel buyback program.

A social planner is now considered who internalizes the negative externality generated by aggregate effort and chooses total effort to maximize social surplus. In contrast to vessels in (2), which maximize private profits taking aggregate effort as given and thus base their decisions on private best responses, the planner internalizes congestion, resource depletion, and spatial spillovers. Absent productivity-enhancing technologies, a reduction in total effort lowers private profits but may raise social welfare if the resulting reduction in the externality is sufficiently large. The necessary condition for an increase in social welfare is

$$\sum_{i=1}^n (b_i^r - c_i^r) + \sum_{i=1}^n \Delta N_i > \sum_{i=1}^n (b_i - c_i), \quad (8)$$

where b_i and c_i denote generalized benefits and costs, and $\Delta N_i = N_i(E_{-i}) - N_i(E_{-i}^r) > 0$ captures the reduction in the negative externality induced by lower aggregate effort. Since $b_i^r - c_i^r < b_i - c_i$ for all vessels, social welfare increases only if the reduction in externalities is sufficiently large to compensate for the decline in private net benefits.

Regulators often lack complete information on vessels' payoff functions and on the magnitude of the externality, which makes direct welfare evaluation difficult. In this context, reduced-form indicators are useful to study how subsidies translate into

⁸Proposition 2 is the only one among the five propositions whose qualitative implications may depend on the payoff symmetry assumption. The remaining propositions are robust to introducing cost heterogeneity across vessels, as they rely only on monotonicity and concavity conditions rather than symmetry.

behavioral responses. The effort-to-subsidy ratio, $\eta_i = \frac{e_i}{S_i}$, provides such an indicator, capturing the pass-through from public expenditure to fishing effort. It measures the amount of effort induced per unit of subsidy, and thus the intensity of policy transmission to externality pressure in the fishery. A decline in η_i is interpreted as an improvement in policy performance, as it reflects lower effort per unit of public spending. Since effort generates negative externalities, lower values of η_i are associated with welfare improvements insofar as they operate through this channel.

Importantly, η_i summarizes the policy-to-effort transmission mechanism but does not measure welfare directly. Under the assumption that effort is monotonically related to externality severity, it provides a sufficient statistic for this transmission channel, although not for welfare itself, which also depends on private net benefits and on the allocation between authorized and unauthorized effort.

Following the reform, changes in η_i capture adjustments in the transmission of subsidies into effort and externality pressure, motivating the following proposition.

Proposition 5. *Without complementary management instruments, a subsidy reduction may increase the authorized effort-to-subsidy ratio, e_{a_i}/S_i , if authorized effort does not fall proportionally. The unauthorized ratio, e_{na_i}/S_i , increases unambiguously due to substitution away from authorized fishing and lower subsidies.*

Proof: See Section 4.5 in the supplementary material.

Consequently, e_{a_i}/S_i is theoretically ambiguous, as it depends on whether authorized effort falls by more or less than the subsidy cut, so its sign cannot be determined ex ante. The associated threshold condition, derived in the proof of Proposition 5, determines when each case occurs based on the costs of authorized fishing, the size of the subsidy reduction, and the marginal benefits from fishing.

Overall, the model presents a set of empirically testable predictions that guide the identification strategy. A reform combining differential subsidy reductions with a VBP is expected to reduce authorized and total effort and may induce exit among less competitive vessels. Higher relative costs of authorized fishing may also incentivize shifts toward unauthorized activities. Additionally, the reform may alter the efficiency with which subsidies translate into fishing pressure, as reflected in the authorized effort-to-subsidy ratio. These implications have direct policy relevance. If subsidy cuts increase unauthorized fishing, complementary monitoring and enforcement may be required. If they primarily affect efficiency, additional management instruments—such as catch shares, effort caps, or spatial closures—may be needed. These predictions guide the empirical analysis in the next section.

4 Methodology

4.1 Data

To examine the relationship between access to fuel subsidies for industrial fishing vessels and fishing activity, we used two main data sources. The first is a newly released database compiled by Wang et al. (2023) that contains individual-level administrative data on fishing vessels registered in the Marine Fishing Vessel Dynamic Management System of the Zhejiang provincial government. This administrative platform comprises five modules corresponding to each section of vessel management activities: engine

power quota, vessel name, vessel inspection, vessel registration, and fishing license. The second is data from Global Fishing Watch (GFW), which provides vessel-level information on fishing activity, measured as daily fishing hours at a spatial resolution of 0.01 degrees per pixel, based on Automatic Identification System (AIS) data.

In this study, we focus on trawler vessels, which are considered one of the most environmentally damaging fishing methods (Buhl-Mortensen and Buhl-Mortensen 2018). The database from Wang et al. (2023) contains reports from 7,592 trawler vessels constructed between 1988 and 2011. A total of 3,354 vessels, representing 44% of the sample, were successfully matched with the GFW data. The resulting database includes fishing activity in hours per day/pixel for 3,354 trawler vessels from Zhejiang province in China for the period 2012-2019. The database also includes characteristics of the vessels, records of fuel subsidy payments received, and records of exits through the VBP.

The descriptive statistics for the outcome variables are presented in Table 1. On average, vessels spend 1.26 hours/day/pixel on fishing activities (3.67 hours/day/pixel conditional on positive fishing hours). In terms of authorization status, vessels averaged 1.25 hours/day/pixel in authorized fishing and 2.79 hours/day/pixel in unauthorized fishing. For positive fishing hours, the averages were 3.67 and 4.27 hours/day/pixel, respectively. No-positive fishing hours refer to pixels where fishing vessels passed through, but did not engage in any fishing activity. This indicates the transit from one fishing location to another; then the statistics presented without zeros refer to the exclusion of records in which only transit occurred, considering only those instances where fishing activity actually took place.

Regarding the probability of exit due to the VBP, on average, 5% of vessels exited the fishing sector after the reform, with smaller vessels showing the highest exit rates (See Figure S2 in the supplementary material).

Table 1 also presents the descriptive statistics before and after the reform⁹. On average, subsidy payments decreased by approximately 45%. In terms of fishing activity, total fishing effort declined by approximately 19% and authorized fishing by about 19%. By contrast, unauthorized fishing increased by around 35%. Notably, the authorization rate remained essentially unchanged, moving from approximately 99.7% to 99.8%. Likewise, the probability of vessel exit increased substantially. Table S1 in the supplementary material reports additional descriptive statistics.

Figure 2 illustrates the extent of the fishing activity of the vessels in the database, along the eastern coast of China. For positive fishing hours, vessels fished an average of 3.67 hours/day/pixel in national territory, 4.90 hours/day/pixel in Taiwanese territory, 4.57 hours in the joint regime area between Japan and Korea, 3.70 hours in Disputed Zones, and 3.67 hours in Japanese territory (Table 2).

⁹It is important to note that the means for authorized and unauthorized fishing hours are calculated as conditional means—restricted to their respective sub-samples of authorization status. Consequently, the table represent the average intensity of effort within each regime rather than an unconditional average over the entire panel, which explains why they are not additive to the total fishing effort.

Table 1: Summary statistics of the main outcome variables before and after reform.

	Total		Before reform		After reform	
	Mean	SD	Mean	SD	Mean	SD
Total fishing hours	1.26	3.06	1.38	3.11	1.12	2.99
Total fishing hours (positive values)	3.67	4.29	3.56	4.15	3.86	4.51
Authorized fishing hours	1.26	3.05	1.37	3.11	1.11	2.98
Authorized fishing hours (positive values)	3.67	4.28	3.55	4.14	3.86	4.50
Unauthorized fishing hours	2.79	4.63	2.46	4.45	3.32	4.86
Unauthorized fishing hours (positive values)	4.27	5.15	4.33	5.17	4.20	5.12
Fuel Subsidies (RMB/Kw)	1,378.47	434.51	1,727.72	93.29	949.78	276.4
Authorization status	0.99	0.05	0.997	0.06	0.998	0.05
Exit probability	0.027	0.16	0.009	0.095	0.05	0.21

Source: Authors, using information from [Global Fishing Watch](#), [Sea Around Us](#) and [Wang et al. \(2023\)](#) database. Note: Statistics are presented for vessels between 25 and 35 meters in length. Means for authorized and unauthorized fishing hours are calculated conditional on their respective authorization status; consequently, they are not additive to total fishing hours due to different denominators (sub-sample sizes) across categories. Fishing hours are measured at the pixel-day level as total hours per pixel per day. Each pixel has a resolution of 0.01 degrees (approximately 1 km $\times$ 1 km). Fuel subsidies are expressed as the annual payment in Chinese Yuan (RMB) per Power (Kw). Authorization status takes the value of 1 if the detected fishing activity carried out by a Chinese vessel occurs in the waters of countries where there is a fishing access agreement, in national waters, or in high sea. Exit probability indicates whether the vessel participates in the VBP.

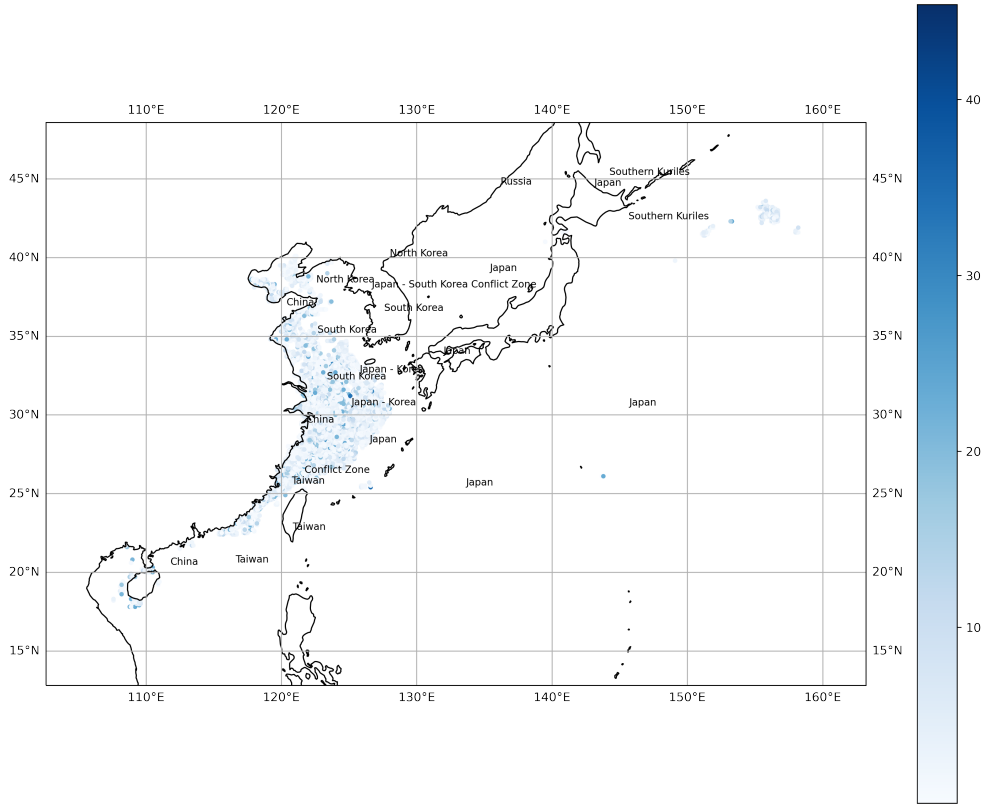


Fig. 2: Coverage of Chinese trawler vessels' fishing activity. Each pixel has a spatial resolution of 0.01 degrees (approximately 1 km $\times$ 1 km). Fishing activity is measured as hours of fishing per vessel per day per pixel. Higher values indicate greater fishing intensity within a given spatial cell and time period. Source: Authors, using information from [Global Fishing Watch](#) and [Wang et al. \(2023\)](#).

Regarding the authorization status of fishing activity, we rely on the Sea Around Us (SAU) database on fishing access agreements. Fishing activity is classified as authorized if a formal agreement exists between the parties. This distinction is necessary because the SAU database includes not only formal agreements but also vessel detection information derived from GFW data, making it essential to clearly differentiate between the types of access reported. Figure S3 in the supplementary material shows the distribution of vessel length by type of fishing activity for vessels between 25 and 35 meters in length, both before and after the reform. Following the reform, the distribution shifted toward a greater share of unauthorized fishing. Table S2 in the supplementary material shows the fishing agreements that China had during the analysis period¹⁰.

Table 2: Chinese fishing activity by countries and sovereign.

Country	Sovereign	Agreements	Fishing Hours > 0		
			Obs	Mean	SD
China	China	-	1,466,122	3.67	4.28
South Korea	South Korea	Yes	1,107	3.95	4.87
Taiwan	Taiwan	No	2,872	4.90	5.46
Disputed Zone	Disputed	No	2,111	3.70	4.60
Japan - Korea	Joint Regime	No	1,198	4.57	5.87
Japan	Japan	No	521	3.99	4.64

Source: Authors, using information from [Global Fishing Watch](#), [Sea Around Us](#), and [Wang et al. \(2023\)](#) database. Note: Positive fishing hours are presented. The agreements were defined according to the internal fishing access agreements dataset from Sea Around Us. The existence of an agreement is determined by whether a formal agreement was signed between the parties for the years under analysis.

4.2 Empirical Strategy

China’s fuel-subsidy reform created two key discontinuities: one based on gross tonnage and engine power ([Englander et al. 2025](#)), and another tied to vessel size for harmful operations such as trawling ([Wang et al. 2023](#)). Our analysis concentrates on the vessel-size threshold because it aligns with our dataset and the policy’s focus, while the first discontinuity does not clearly separate binding from non-binding vessels.

This redesign motivates the implementation of a RDD, notably, vessels just over 30 meters in length received approximately 19% more in subsidy payments than those just under 30 meters. Building on this discontinuity, we implement a RDD using vessel length as the running variable and a cutoff at 30 meters. Treatment is deterministically assigned to vessels with length equal to or above this threshold, which experienced differential changes in subsidy intensity following the reform.

¹⁰For agreements with unspecified end year, we assume that the agreement remains valid. This constitutes a conservative coding choice that may affect the classification of authorization status.

We estimate local treatment effects using the nonparametric framework of [Calonico et al. \(2014\)](#), relying on local linear regressions estimated on either side of the cut-off. The parameter of interest is the discontinuity in unauthorized fishing effort at the threshold. Estimation employs data-driven optimal bandwidth selection and bias-corrected confidence intervals. Identification rests on the continuity of potential outcomes at the cutoff, which we assess using standard RDD diagnostics, including covariate balance and density tests around the threshold. Standard errors are clustered at the vessel level.

However, the reform affected not only vessels around the cutoff, but also the broader distribution of the fleet. While the magnitude of subsidy reductions varied across vessels, all were exposed to changes in subsidy intensity. As a result, the RDD identifies local effects at the threshold, whereas a DiD approach allows us to exploit variation across the full sample and obtain more generalizable estimates. Accordingly, we complement the RDD analysis with a DiD specification that differentiates vessels by size in line with the program’s design, restricting the sample to vessels between 25 and 35 meters in length.

The proposed model aims to estimate the effect of differential reduction in subsidy allocation on fishing activity and the number of unauthorized fishing hours, using the following specification

$$y_{izt} = \alpha_1 D_i + \alpha_2 Post_{t_y} + \beta D_i \times Post_{t_y} + \gamma_i + \phi_z + \rho_t + \epsilon_{izt}. \quad (9)$$

Where y_{izt} is the unauthorized fishing efforts for vessel i at pixel z on the day t . The variable $D_i = \mathbf{1}\{\text{Vessel Length} \geq 30\}$ indicates whether the vessel has a length greater than 30 meters. This is a threshold chosen based on the evaluation of subsidy access distribution by vessel size (see Figure S4 and S5 in the supplementary material). $Post_{t_y}$ takes the value of 1 from year $t_y = 2016$, when the subsidy program reform was implemented, and 0 otherwise. The idiosyncratic error term is denoted by ϵ_{izt} , and standard errors are clustered at the vessel level. The coefficient of interest is β which captures the effect of the reform to the fuel subsidy program by large vessels.

Our empirical specification incorporates a rich set of fixed effects to flexibly control for unobserved heterogeneity across multiple dimensions. Vessel fixed effects (γ_i) absorb all time-invariant characteristics of each vessel, such as size, technology, and baseline fishing capacity. Temporal fixed effects (ρ_t) include year, month, and their interactions (year-by-month), allowing us to control for aggregate shocks, seasonality, and time-specific fluctuations in fishing activity, including policy changes and environmental conditions that vary at high frequency. In addition, we include spatial fixed effects (ϕ_z) based on grid-cell latitude and longitude, as well as EEZ identifiers, to account for persistent spatial heterogeneity in fishing conditions, regulatory environments, and resource availability. Together, this high-dimensional fixed effects structure ensures that identification comes from within-vessel variation over time, net of flexible temporal and spatial confounders.

To avoid potential bias from manipulation of the treatment assignment rule, the database consists of vessels built before 2012. Given this, and considering that D_i is a characteristic that does not vary over time in the analysis period, there is no reason

to believe that D_i is correlated with unobserved factors that determine the outcome variable. Therefore, it is not necessary to include additional covariates.

Taking into account the distribution of vessels by size, the regressions are estimated for vessels around the 30-meter threshold. As the main estimation, the regressions are conducted for vessels between 25 and 35 meters.

Based on equation (9), we made the following change: similar to Wang et al. (2023), we replaced $\beta D_i \times Post_{t_y}$ with $\beta_j D_i \times Post_{t_y}^j$ for $j = 2012, \dots, 2019$, where $Post_{t_y}^j = \mathbb{1}\{t = j\}$. This approach allows us to estimate the treatment effect over time by conducting an event study that captures the impact of the subsidy reduction resulting from the program reform. We use the following specification

$$y_{izt} = \alpha_1 D_i + \alpha_2 Post_{t_y} + \sum_{j=2013}^{2019} \beta_j D_i \times Post_{t_y}^j + \gamma_i + \phi_z + \rho_t + \epsilon_{izt}. \quad (10)$$

To ensure correct estimation of the causal effect of the change in subsidy access on unauthorized fishing activity using the DiD method, it is necessary to satisfy the following assumptions: (i) *No anticipation*. This refers to the assumption that fishers did not have the opportunity to adjust vessel characteristics to take advantage of the treatment assignment rule. To prevent any bias, only vessels built before 2012 are considered. (ii) *Parallel trends*. This assumption presupposes that the fishing behavior of vessels that had access to more subsidies and those that had access to fewer subsidies follows parallel trends before the reform. (iii) *No interference between units (SUTVA)*. It is possible that this assumption is violated, which calls for a cautious interpretation of the results. The estimation captures the difference between the impact on the treatment group and the control group; however, because the control group is also affected by the treatment, the results differ from what would have been observed if the treatment group had not been exposed to the reform at all. While this approach still allows us to identify the relative differential effect of the reform across the defined groups, the results should be interpreted as a *relative effect* rather than the absolute impact of the treatment compared to a purely untreated counterfactual scenario.

5 Results

This section proceeds in two parts. Section 5.1 presents descriptive evidence on the evolution and composition of fishing effort before and after the reform, documenting key patterns in the data that are consistent with the model's predictions. Section 5.2 then provides causal evidence using the empirical strategies described above, and further explores the underlying mechanisms by analyzing heterogeneity across vessel types and adjustments in effort relative to subsidy changes.

5.1 Aggregate Trends and Descriptive Evidence

We begin our analysis by documenting the aggregate shifts in fishing activity following the 2016 reform. The policy change led to a drastic reduction in financial support, with average fuel subsidy payments falling by approximately 69% (see Figure S5 in

the supplementary material). This cost shock, coupled with the VBP, induced a significant restructuring of the fleet: smaller vessels, which faced proportionally larger subsidy cuts, exhibited a 2.5% higher probability of exiting the sector compared to larger vessels (see Figure S2 in the supplementary material). These patterns provide support for Proposition 4, suggesting that subsidy reductions effectively incentivize the retirement of less competitive fishing capacity.

5.1.1 Total Effort and the Reallocation of Fishing Activity

Figure 3 illustrates the evolution of total fishing hours for both treated (large vessels, ≥ 30 m) and control (small vessels, < 30 m) vessels. We observe that the reform led to a contraction in aggregate fishing pressure. While both groups exhibit a downward trend after 2016, the decline is more pronounced among smaller vessels, consistent with their greater sensitivity to cost increases and higher exit rates, in line with Proposition 3.

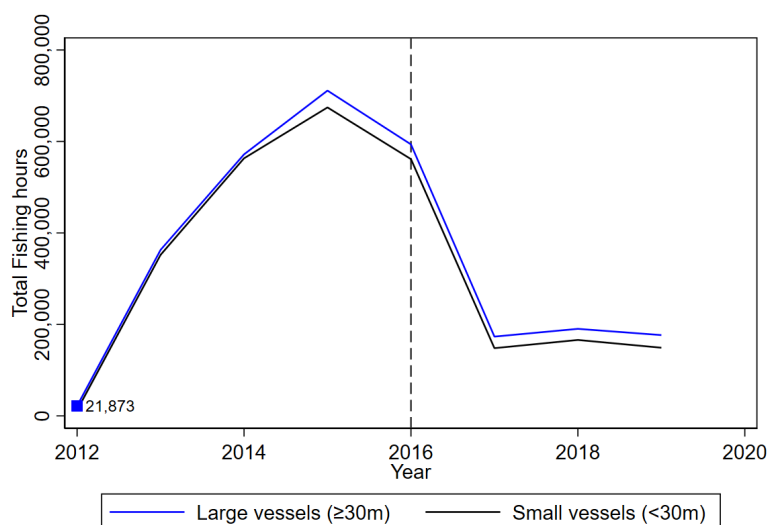


Fig. 3: Total fishing effort by treatment group (2012–2019). The figure plots the annual total fishing hours for vessels between 25 and 35 meters in length, distinguishing between treated vessels ($\geq 30m$, larger subsidy reductions) and control vessels ($< 30m$). The vertical dashed line marks the 2016 fuel subsidy reform. Fishing effort is measured as the sum of AIS-based fishing hours from GFW. Values in early years (e.g., 2012) may appear close to zero due to scale differences rather than absence of activity. Source: Authors, using information from [Global Fishing Watch](#) and [Wang et al. \(2023\)](#) database.

However, the aggregate decline masks a crucial shift in the composition of fishing effort. Figure 4 disaggregates total fishing hours by authorization status. As predicted by Proposition 1, authorized fishing effort experienced a sharp and sustained decline immediately after the vertical dashed line marking the reform. In contrast, unauthorized fishing hours showed an important increase. This descriptive evidence suggests

an *effort leakage* or substitution effect: as the effective marginal cost of authorized fishing rose, a subset of the fleet reallocated effort toward unauthorized areas, validating the core mechanism of Proposition 2, although the substitution remains incomplete, as total fishing effort does not return to its pre-reform levels.

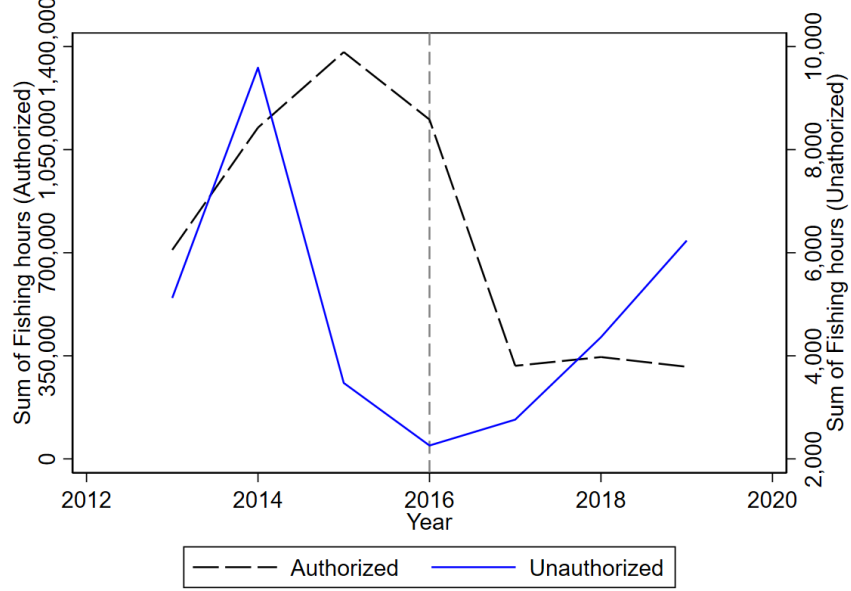


Fig. 4: Fishing effort by authorization status (2013–2019). The figure shows total annual fishing hours disaggregated into authorized and unauthorized activity for vessels between 25 and 35 meters. Data start in 2013 because no unauthorized fishing observations are recorded in 2012. The vertical dashed line indicates the 2016 reform. While authorized fishing declines after the reform, unauthorized fishing increases, suggesting a reallocation of effort. Source: Authors, using information from [Global Fishing Watch](#) and [Wang et al. \(2023\)](#) database.

5.1.2 Heterogeneity and Spatial Shifts

To understand which vessels are driving this shift, Figure 5 displays the share of unauthorized fishing effort by vessel size. Prior to the reform, smaller vessels often exhibited higher shares of unauthorized activity, likely supported by high subsidy intensity. Post-2016, the pattern reverses: larger vessels ($\geq 30m$) significantly increase their share of unauthorized fishing, while smaller vessels reduce theirs. This suggests that larger vessels, with greater physical capacity for long-distance trips, were better positioned to exploit unauthorized fishing opportunities to offset the loss of domestic subsidies.

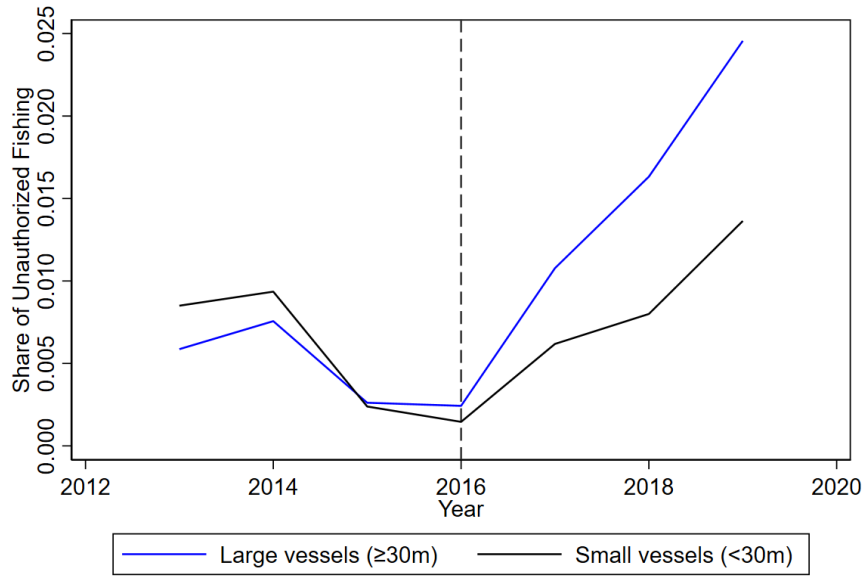
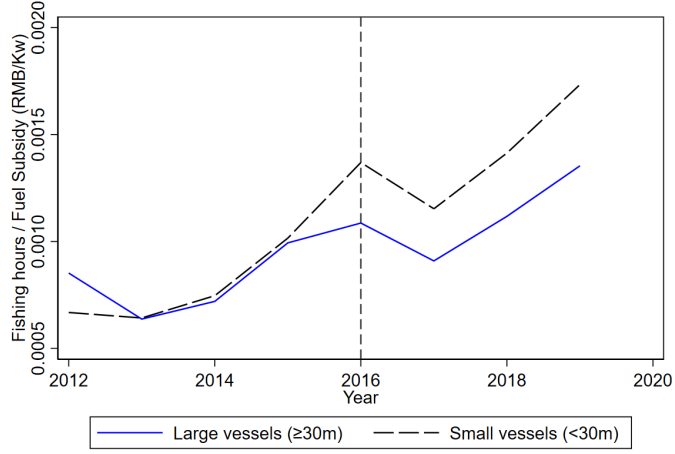


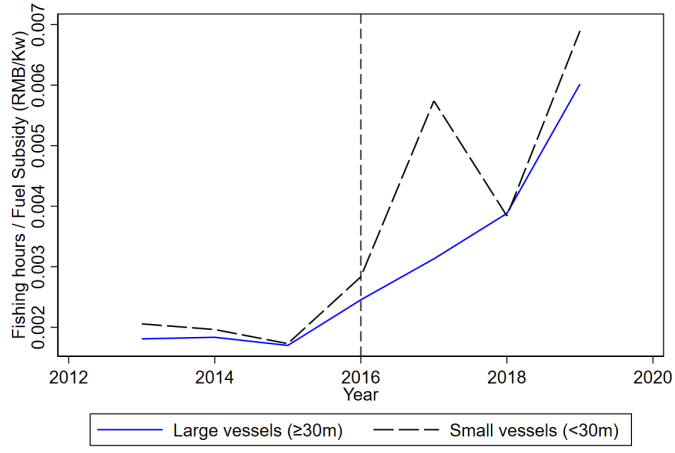
Fig. 5: Share of unauthorized fishing effort by vessel size (2013–2019). The figure presents share of unauthorized fishing hours (unauthorized over total fishing effort) by year, distinguishing vessels above and below the 30-meter threshold. The sample includes vessels between 25 and 35 meters. Data start in 2013 because no unauthorized fishing observations are recorded in 2012. The figure shows a post-reform increase in unauthorized fishing driven primarily by larger vessels. Source: Authors, using information from [Global Fishing Watch](#) and [Wang et al. \(2023\)](#) database.

5.1.3 Compensatory Behavior and Subsidy Efficiency

Finally, we examine how the relationship between subsidies and effort changed for those vessels that remained in the fishery. Proposition 5 suggests that if effort does not fall proportionally to the subsidy budget, the *efficiency* of each unit of subsidy in generating fishing pressure actually increases.



(a) Authorized Fishing Efforts



(b) Unauthorized Fishing Efforts

Fig. 6: Fishing effort and fuel subsidy payments by vessel length (2013–2019). The figure presents annual fishing hours (AIS-based) and fuel subsidy payments (RMB per kW) for vessels between 25 and 35 meters, distinguishing vessels above and below the 30-meter threshold. Panel (a) shows authorized fishing effort, while panel (b) shows unauthorized fishing effort. The vertical dashed line indicates the 2016 fuel subsidy reform. Data start in 2013 because no unauthorized fishing observations are recorded in 2012. The figure illustrates differential post-reform adjustments in fishing behavior across vessel sizes, with a relative increase in unauthorized fishing among larger vessels. Source: Authors, using information from [Global Fishing Watch](#) and [Wang et al. \(2023\)](#) database.

Figure 6 provides evidence of this adjustment. The ratio of fishing hours to fuel subsidy payments (RMB/kw) rose for both authorized (Panel a) and unauthorized (Panel b) fishing after 2016. This upward trend indicates a compensatory response

along the intensive margin: fishers who did not exit the sector intensified their activity per unit of financial support received. This increase in the effort-to-subsidy ratio highlights a potential policy consideration: while the reform successfully reduced the total aggregate effort, it also increased the intensity of fishing pressure per unit of subsidy.

5.2 Causal Effects of the Reform

To establish a causal link between subsidy reductions and changes in fishing behavior, we employ two complementary econometric strategies. First, a RDD exploits the discrete jump in subsidy intensity at the 30-meter vessel length threshold. Second, a DiD approach captures the average treatment effect across the broader fleet, accounting for vessel, time and spatial fixed effects.

5.2.1 Local Treatment Effects: RDD Evidence

The RDD results confirm that the 2016 reform created a significant discontinuity in financial support. As shown in Table 3, vessels just above the 30-meter threshold received between 22.8% and 32.1% more in fuel subsidies (in logs) compared to those just below the cutoff. This jump is visually evident in the top panels of Figure 7, where a clear gap emerges post-reform, while no such discontinuity existed in the pre-reform period.

Table 3: Regression discontinuity estimates of fuel subsidies and unauthorized fishing at the 30-meter threshold.

	Fuel Subsidies (log)			Unauthorized Fishing (log hours)		
	2016		2018	2016		2018
	Optimal BW	Fixed BW	Optimal BW	Optimal BW	Fixed BW	Optimal BW
RDD Estimate	0.321*** (0.045)	0.228*** (0.033)	0.223*** (0.078)	0.018 (0.290)	0.238 (0.192)	-0.359 (0.458)
Observations	863,710	863,710	381,434	783	783	1,527
Bandwidth	1.36	5.00	1.17	1.17	5.00	0.80
						-0.535** (0.218)
						1,527
						5.00

Source: Authors, using information from [Global Fishing Watch](#) and [Wang et al. \(2023\)](#) database.

Note: The table reports RDD estimates at the 30-meter vessel length threshold. Outcomes include log fuel subsidies (RMB/kW) and log unauthorized fishing hours. Estimates correspond to bias-corrected local polynomial regressions with robust standard errors clustered at the vessel (MMSI) level, following [Calonico et al. \(2014\)](#).

Two bandwidth choices are reported: the mean squared error (MSE)-optimal bandwidth (optimal BW) and a fixed bandwidth (Fixed BW) of 5 meters. The 2016 estimates capture the immediate effects of the reform, while the 2018 estimates reflect a more consolidated policy environment after subsidy reductions and partial convergence across vessel sizes.

The RDD identifies local differences in outcomes associated with marginal changes in subsidy intensity at the cutoff.

Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

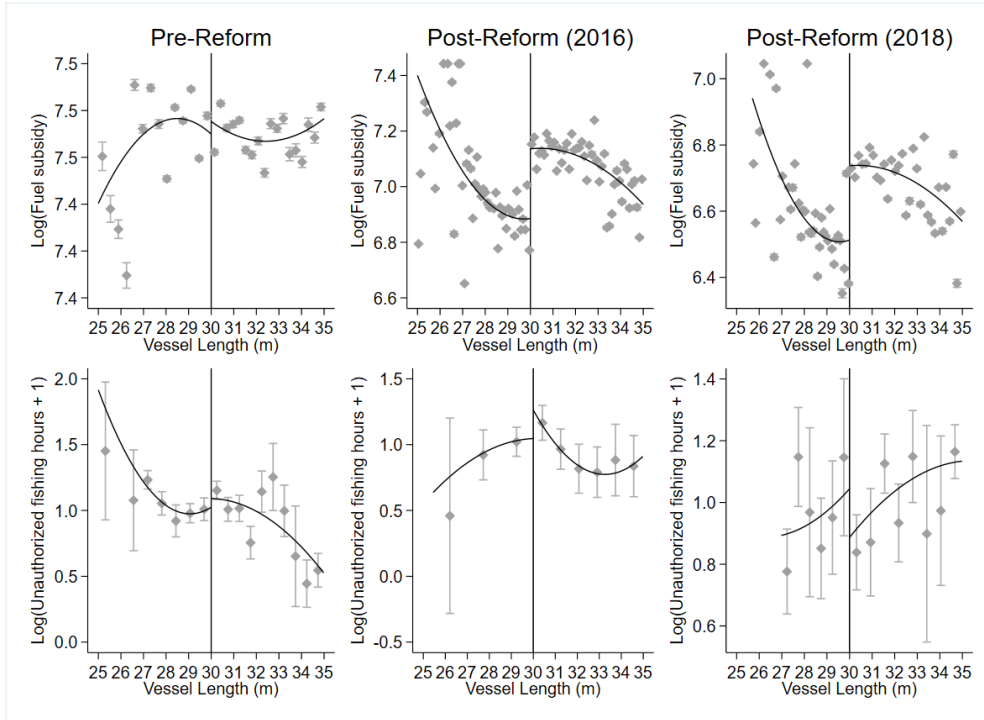


Fig. 7: Regression discontinuity estimates at the 30-meter vessel length threshold. This figure presents RDD plots for the logarithm of fuel subsidies (top panel) and the logarithm of unauthorized fishing hours (bottom panel) as a function of vessel length, before (left) and after (right). The vertical line indicates the 30-meter cutoff that determines eligibility for the fuel subsidy program. Observations are grouped into bins, and local polynomial regressions are fitted separately on each side of the threshold. The running variable is vessel length (in meters). Source: Authors, using information from [Global Fishing Watch](#) and [Wang et al. \(2023\)](#) database.

This differential shock in subsidies translated into immediate shifts in unauthorized fishing activity. In 2016, the RDD estimate for unauthorized fishing hours yields a positive point estimate (0.238 in the fixed bandwidth specification), while this coefficient does not reach statistical significance at the 30-meter cutoff, its positive sign indicates that larger vessels—which retained a higher subsidy intensity—increased their presence in unauthorized waters relative to smaller vessels. Additionally, we evaluate the impact of the reform on the extensive margin of fishing activity. Figure S6 and Table S3 in the supplementary material shows that the jump in subsidy intensity also significantly increased the probability of participating in unauthorized fishing for vessels just above the threshold in 2016, mirroring the intensive margin results found for fishing hours.

However, we find evidence of a temporal reversal. By 2018, the RDD estimate for unauthorized fishing becomes negative and significant (Table 3). This suggests that over time, smaller vessels (the *control* group at the margin) adjusted their behavior along the intensive margin, increasing their unauthorized effort per unit of subsidy to compensate for the larger cuts they faced. This reversal is consistent with Propositions

2 and 5, and the gradual adjustment process: while the initial shock favored larger vessels, the continued decline in subsidies eventually pushed smaller vessels to intensify their effort to maintain competitiveness.

The validity and robustness of these RDD estimates are further supported by a comprehensive set of diagnostic tests provided in the Supplementary Material. First, Figure S7 demonstrates that predetermined vessel characteristics—including engine power, vessel age, net tonnage, and vessel breadth—remain continuous at the 30-meter threshold. This confirms that vessels just above and below the cutoff are comparable in observable dimensions, mitigating concerns regarding systematic differences between the treatment and control groups at the margin. Second, a placebo threshold test; Figure S8 confirms that the estimated discontinuities are unique to the 30-meter policy boundary and do not emerge at alternative, non-regulatory cutoffs. Finally, we examine the density of vessel lengths around the threshold to address potential manipulation of the running variable; Figure S9 shows no evidence of bunching or discontinuities in the distribution of vessels, effectively ruling out strategic sorting by fishers around the eligibility cutoff.

5.2.2 Average Treatment Effects: DiD and Event Study

While the RDD identifies local effects at the cutoff, the DiD specification provides a more generalizable estimate of the reform’s impact on the Zhejiang fleet. Table 4 presents the average treatment effect on unauthorized fishing hours. Our preferred specification (Column 4), which includes vessel, month-year, and pixel-level fixed effects, indicates that unauthorized fishing effort increased by 19.3% per each spatial unit (pixel-day) among larger vessels following the reform¹¹. Placebo tests (Table S4 in the supplementary material) show no consistent pre-trends, supporting the validity of the identification strategy.

This result provides strong empirical validation for Proposition 2. The reduction in fuel subsidies increased the effective marginal cost of authorized fishing, thereby reducing its relative profit advantage. As a result, vessels reallocated their effort toward unauthorized zones—particularly in disputed maritime areas—where the expected returns remained high enough to offset the increased operating costs (see Figure S10 and Figure S11 in the supplementary material).

¹¹This result is relative to the control group. As discussed in Section 4, this coefficient identifies the differential response between vessel types, capturing the divergence in incentives generated by the asymmetric subsidy cuts. While potential violations of SUTVA in this setting may complicate the interpretation of absolute counterfactual outcomes, they do not invalidate the identification of the policy-induced behavioral wedge across fleet segments.

Table 4: Average treatment effect of fuel subsidy program reform on unauthorized fishing activity

	Unauthorized fishing hours (<i>log</i>)			
	(1)	(2)	(3)	(4)
1.Treated*1.Post	0.018 (0.066)	0.201** (0.089)	0.206** (0.080)	0.193** (0.079)
Vessel FE	No	Yes	Yes	Yes
Year & Month FE	No	No	Yes	Yes
Pixel FE	No	No	No	Yes
Mean _{<2016} Log(y_{izt})	0.993	0.993	0.993	0.993
N Treated	5,902	4,284	4,282	4,266
N Control	4,324	5,839	5,838	5,810
Observations	10,226	10,123	10,120	10,076

Source: Authors, using information from [Global Fishing Watch](#) and [Wang et al. \(2023\)](#) database.
Note: * $p < .10$, ** $p < .05$, *** $p < .01$. The outcome variables are expressed in log. Vessel, year, month and pixel Fixed Effects are included. Standard errors clustered at the vessel level are reported in parentheses. The mean represents the average of the outcome variable before the reform for the control group.

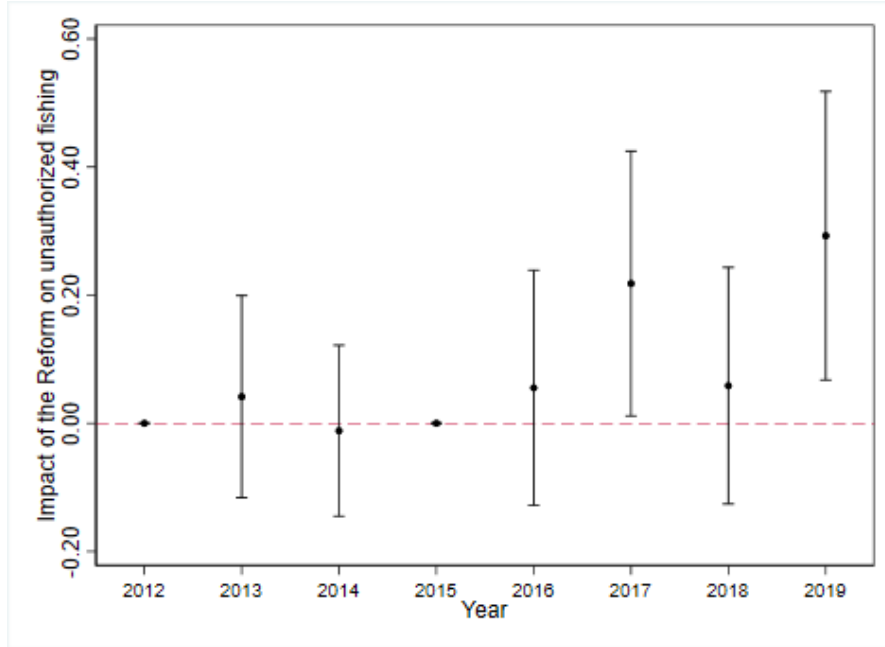


Fig. 8: Event study of fuel subsidy program reform effect on unauthorized fishing hours. The vertical bars indicate the confidence intervals at 10%. The figure presents the results of the estimates based on equation (10). The outcome variables are expressed in log. Vessel, year, month and pixel fixed effects are included. Standard errors are clustered at the vessel level. Source: Authors, using information from [Global Fishing Watch](#) and [Wang et al. \(2023\)](#) database.

To validate the causal interpretation, the event study in Figure 8 confirms the absence of pre-existing trends. Prior to 2016, the estimated coefficients for the interaction between vessel size and year are not statistically significant. Following the reform, we observe a significant and persistent increase in unauthorized fishing that emerges in 2017 and continues through 2019. This lag in the response suggests that fishers required time to reorganize their operations and develop the necessary logistics for more frequent or distant unauthorized trips. Additionally, we explored continuous measures of subsidy variation; however, estimates were imprecise and sensitive to specification, likely reflecting nonlinear responses and measurement noise. This reinforces our focus on the policy-induced discontinuity as the primary source of identification.

The combination of subsidy reductions and the VBP program increased the probability of exit among smaller vessels by 2.5% relative to larger vessels. However, the magnitude of this effect is moderate: a 1% reduction in subsidies increases the probability of exit by only 0.16 percentage points (Table S5 in the supplementary material). This suggests gradual adjustment rather than abrupt restructuring.

In summary, these findings suggest a two-stage adjustment process. In the short run, larger vessels respond more strongly to relative incentives and expand unauthorized fishing. In the medium run, as subsidies continue to decline and differences across vessel sizes narrow, smaller vessels increasingly adjust along the intensive margin, raising effort per unit of subsidy. This dynamic helps reconcile the DiD and RDD results and highlights the importance of distinguishing between aggregate and local treatment effects, as well as short- and medium-term responses.

6 Discussion and Concluding Remarks

This paper provides causal evidence of how a significant reduction in fuel subsidies—traditionally considered *capacity-enhancing*—reshapes the behavior of one of the world’s largest fishing fleets. While the 2016 Chinese reform successfully reduced aggregate fishing effort, our results reveal a critical and unintended side effect: a 19.3% increase in unauthorized fishing activity, particularly in geopolitically sensitive and disputed zones. This finding suggests that subsidy reforms, in the absence of robust monitoring and enforcement, may induce an *effort leakage* that undermines international conservation goals and exacerbates maritime tensions.

Our main contribution is the documentation of the reallocation of fishing effort across space and legal regimes. From a theoretical perspective, the reform altered the relative cost advantage of authorized versus unauthorized fishing. For larger vessels, the retention of higher subsidy levels relative to smaller ones provided the financial liquidity and physical capacity to expand operations into foreign waters where China lacks formal access agreements. This expansion is consistent with the argument that capacity-enhancing subsidies enable longer and more distant trips, effectively reducing the private cost of long-distance fishing (Englander et al. 2025; Finley 2017).

In contrast, smaller vessels faced a double pressure: larger proportional subsidy cuts and increased competition from larger vessels. While many exited the sector via the VBP—validating the role of exit incentives in reducing overcapacity (Wang et al. 2023)—those that remained exhibited a *compensatory response*. By increasing

their effort-to-subsidy ratio, these vessels intensified their activity to mitigate the loss of competitiveness. This suggests that subsidy reductions do not translate into a proportional decline in fishing pressure. Instead, they may trigger a reallocation of effort—a phenomenon we define as *effort leakage*—whereby domestic conservation gains are partially offset by increased pressure in international or disputed zones.

Our results provide quantitative evidence that domestic support programs have far-reaching geopolitical consequences (Zhang 2016; Midford and Østhagen 2024). The marked increase in Chinese vessel presence in the waters of Taiwan, Japan, and Disputed Zones highlights how national subsidy policies can interfere with international maritime stability. By reducing the cost of operating in contested areas, fuel subsidies may inadvertently facilitate *gray-zone* activities that challenge international norms of maritime jurisdiction.

This study aligns with the definitions of the FAO (2024) to categorize unauthorized fishing as a critical dimension of IUU fishing. Our findings also support the broader debate that input subsidies can facilitate activities inconsistent with global sustainability goals (Sumaila et al. 2021, 2010b). When domestic instruments reduce the marginal cost of fuel, they distort fishers’ incentives to respect maritime boundaries, especially in contexts where institutional enforcement is weak or geopolitically sensitive.

This interaction between domestic subsidy policy and international maritime security underscores the importance of the WTO negotiations on fisheries subsidies (SDG 14.6). While the WTO Agreement on Fisheries Subsidies aims to eliminate support for IUU fishing, our study shows that even broad reforms targeting *overcapacity* can inadvertently increase IUU-like behavior if the underlying incentive structure pushes vessels toward contested waters. Therefore, subsidy discipline must be coupled with enhanced transparency and regional cooperation in monitoring, control, and surveillance (MCS).

The Chinese experience offers a cautionary tale for other distant-water fishing nations. A *subsidy-only* approach to fisheries management is likely to be insufficient. Our analysis suggests three key policy lessons: i) The VBP acted as a crucial *safety valve* by providing a retirement path for less competitive vessels. Without such a program, the shift toward unauthorized fishing could have been even more pronounced as fishers struggled to cover rising costs. Future reforms should ensure that exit bonuses are sufficiently high to outweigh the expected returns of illegal activity. ii) Subsidy reductions are a price-based instrument. For them to work as intended, the *price* of illegal activity (e.g. probability of detection multiplied by the fine) must also be high. The increase in unauthorized fishing suggests that the reform’s *push* factor was stronger than the current *deterrence* factor in foreign waters. And iii) since fishers respond by increasing the intensity of effort per unit of remaining subsidy, regulators must account for these non-linear behavioral adjustments. Policy effectiveness should be measured not just by the reduction in fiscal expenditure, but by the net change in total fishing pressure across all jurisdictions.

Despite the robust causal framework, this study has limitations that open avenues for future research. Our reliance on AIS data from GFW means we only capture vessels that broadcast their positions. While GFW uses sophisticated cross-referencing to detect location manipulation (Paolo et al. 2024), unobserved activity remains a

concern, particularly for vessels that deliberately disable transponders when entering unauthorized zones. Furthermore, our dataset is restricted to official subsidy recipients (Wang et al. 2023), potentially understating the total scale of unauthorized activity by non-compliant or *dark* vessels. Future research should incorporate data from other coastal provinces to assess the national scale of these behavioral adjustments. Finally, although our empirical strategies exploit quasi-experimental variation through both DiD and RDD—and are supported by the absence of pre-trends and continuity tests—remaining concerns may arise from concurrent changes around the reform or near the policy threshold that could differentially affect vessel behavior.

In conclusion, while China’s fuel subsidy reform is a step in the right direction toward meeting international commitments, it highlights the complexity of managing common-pool resources. Reducing subsidies is a necessary but not sufficient condition for sustainable fisheries. To prevent the displacement of fishing effort and the escalation of maritime conflicts, nations must integrate economic reforms with robust international cooperation and rigorous enforcement mechanisms. The path to *Blue Transformation* requires a policy mix that aligns domestic economic incentives with the global reality of shared marine ecosystems.

Data, Materials, and Software Availability. The data and codes are deposited in a Zenodo repository (<https://doi.org/10.5281/zenodo.17256749>). This material includes the datasets used as well as the codes required to replicate the figures and tables presented in the document.

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Credit authorship contribution statement. Jorge Montero contributed to conceptualization, methodology, formal analysis and investigation, writing – original draft preparation, writing – review and editing, resources, and supervision. Jorge Marco contributed to conceptualization, writing – original draft preparation, writing – review and editing, and supervision.

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